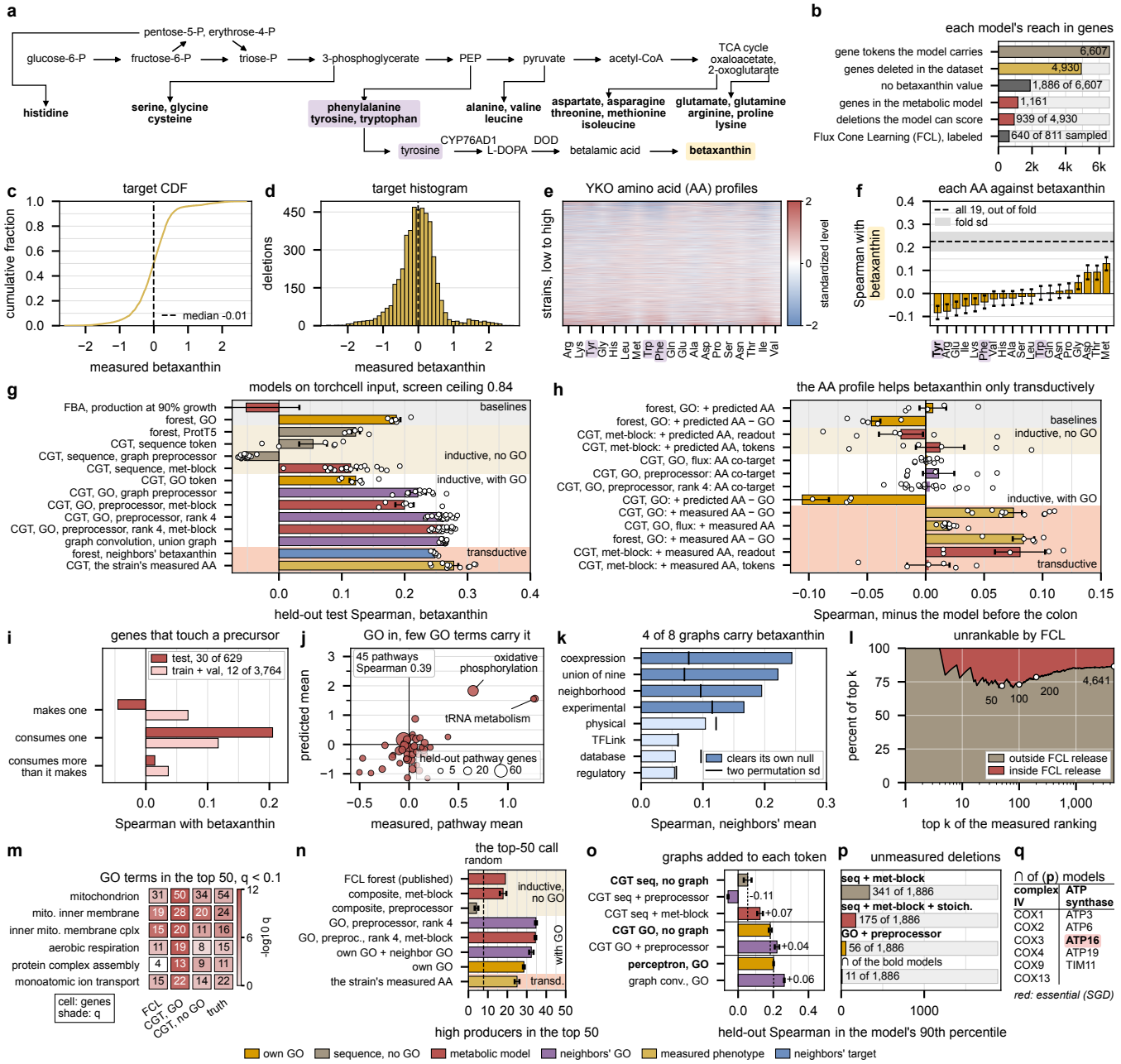

a
pentose-5-P, erythrose-4-P
glucose-6-P
fructose-6-P
triose-P
3-phosphoglycerate
PEP
pyruvate
acetyl-CoA
TCA cycle
oxaloacetate,
2-oxoglutarate
histidine
serine, glycine
cysteine
phenylalanine
tyrosine, tryptophan
alanine, valine
leucine
aspartate, asparagine
threonine, methionine
isoleucine
glutamate, glutamine
arginine, proline
lysine
tyrosine
CYP76AD1
L-DOPA
DOD
betalamic acid
betaxanthin
b
each model's reach in genes
gene tokens the model carries
6,607
genes deleted in the dataset
4,930
no betaxanthin value
1,886 of 6,607
genes in the metabolic model
1,161
deletions the model can score
939 of 4,930
Flux Cone Learning (FCL), labeled
640 of 811 sampled
0
2k
4k
6k
c
target CDF
cumulative fraction
median -0.01
measured betaxanthin
d
target histogram
deletions
measured betaxanthin
e
YKO amino acid (AA) profiles
strains, low to high
standardized level
Arg
Lys
Tyr
Gly
His
Leu
Met
Trp
Phe
Gln
Glu
Ala
Asp
Pro
Ser
Asn
Thr
Ile
Val
f
each AA against betaxanthin
Spearman with betaxanthin
all 19, out of fold
fold sd
Tyr
Arg
Glu
Ile
Lys
Val
His
Ala
Ser
Leu
Trp
Gln
Asn
Pro
Gly
Asp
Thr
Met
g
models on torchcell input, screen ceiling 0.84
FBA, production at 90% growth
forest, GO
forest, ProtT5
CGT, sequence token
CGT, sequence, graph preprocessor
CGT, sequence, met-block
CGT, GO token
CGT, GO, graph preprocessor
CGT, GO, preprocessor, met-block
CGT, GO, preprocessor, rank 4
CGT, GO, preprocessor, rank 4, met-block
graph convolution, union graph
forest, neighbors' betaxanthin
CGT, the strain's measured AA
baselines
inductive, no GO
inductive, with GO
transductive
held-out test Spearman, betaxanthin
h
the AA profile helps betaxanthin only transductively
forest, GO: + predicted AA
forest, GO: + predicted AA − GO
CGT, met-block: + predicted AA, readout
CGT, met-block: + predicted AA, tokens
CGT, GO, flux: AA co-target
CGT, GO, preprocessor: AA co-target
CGT, GO, preprocessor, rank 4: AA co-target
CGT, GO: + predicted AA − GO
CGT, GO: + measured AA − GO
CGT, GO, flux: + measured AA
forest, GO: + measured AA − GO
CGT, met-block: + measured AA, readout
CGT, met-block: + measured AA, tokens
baselines
inductive, no GO
inductive, with GO
transductive
Spearman, minus the model before the colon
i
genes that touch a precursor
test, 30 of 629
train + val, 12 of 3,764
makes one
consumes one
consumes more
than it makes
Spearman with betaxanthin
j
GO in, few GO terms carry it
45 pathways
Spearman 0.39
oxidative
phosphorylation
tRNA metabolism
held-out pathway genes
5
20
60
predicted mean
measured, pathway mean
k
4 of 8 graphs carry betaxanthin
coexpression
union of nine
neighborhood
experimental
physical
TFLink
database
regulatory
clears its own null
two permutation sd
Spearman, neighbors' mean
l
unrankable by FCL
percent of top k
4,641
50
100
200
outside FCL release
inside FCL release
top k of the measured ranking
m
GO terms in the top 50, q < 0.1
mitochondrion
mito. inner membrane
inner mito. membrane cplx
aerobic respiration
protein complex assembly
monoatomic ion transport
-log10 q
cell: genes
shade: q
FCL
CGT, GO
CGT, no GO
truth
n
the top-50 call
random
FCL forest (published)
composite, met-block
composite, preprocessor
GO, preprocessor, rank 4
GO, preproc., rank 4, met-block
own GO + neighbor GO
own GO
the strain's measured AA
inductive,
no GO
transd.
with GO
high producers in the top 50
o
graphs added to each token
CGT seq, no graph
CGT seq + preprocessor
CGT seq + met-block
CGT GO, no graph
CGT GO + preprocessor
perceptron, GO
graph conv., GO
-0.11
+0.07
+0.04
+0.06
held-out Spearman in the model's 90th percentile
p
unmeasured deletions
seq + met-block
341 of 1,886
seq + met-block + stoich.
175 of 1,886
GO + preprocessor
56 of 1,886
∩ of the bold models
11 of 1,886
q
∩ of (p) models
complex IV
ATP synthase
COX1
COX2
COX3
COX4
COX9
COX13
ATP3
ATP6
ATP16
ATP19
TIM11
red: essential (SGD)
own GO
sequence, no GO
metabolic model
neighbors' GO
measured phenotype
neighbors' target